ROBUST NORMAL-ONLY INDUSTRIAL ANOMALY DETECTION

Detection · localization · capture shift · calibration · selective human review

12 categories · 10,821 images · 54 tests

01 · QUESTION & PROTOCOL

Can a simple normal-only system remain useful when capture conditions change—and can it expose when a human should review?

VisA official one-class split

7,358 training-normal · 1,301 validation-normal · 2,162 official test (1,200 anomaly / 962 normal). Duplicate groups cannot cross splits.

Primary system: frozen ResNet-18 multi-scale features → bounded k-center memory (512/category) → conformal p-value → normal / review / anomalous.

02 · CLEAN RESULT

0.8674
IMAGE AUROC

0.9557
PIXEL AUROC

0.7762
AUPRO@30

0.0430
CLEAN NORMAL FPR

~60% less category-index storage than the 2,048-entry reference, with stronger image ranking and region overlap.

03 · WHAT FAILED

0.8686
SEVERE BLUR FPR

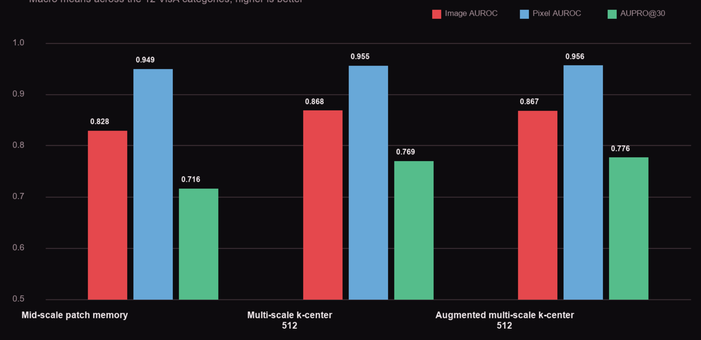
0.5783
UNSEEN-CATEGORY AUROC

- Blur/noise/JPEG create confident nuisance false positives.
- Geometric heatmap correlation falls to 0.442.
- Photometric shift can increase anomaly misses.
- Clean conformal control is not a shift guarantee.

04 · MODEL PROGRESSION

Clean detection and localization

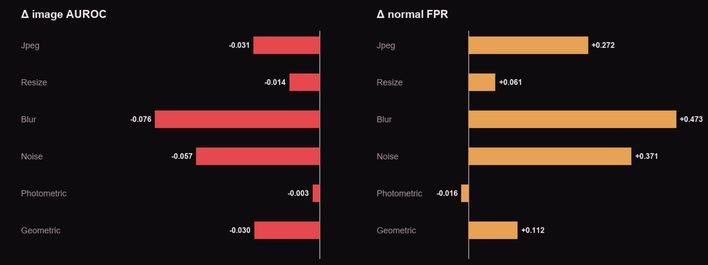
Macro means across the 12 VisA categories; higher is better



05 · ROBUSTNESS

Robustness degradation by shift family

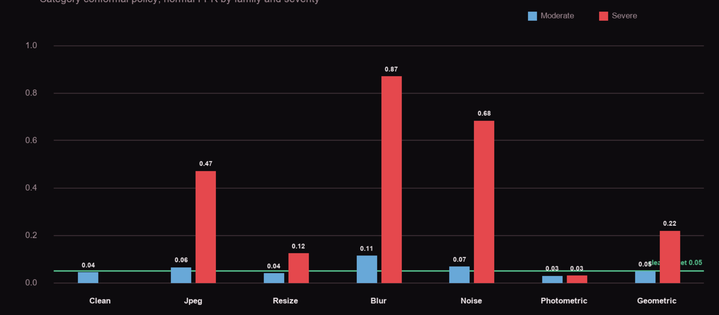
Primary augmented coreset; means across two severities and 12 categories



06 · CALIBRATION BREAKS

False positives under capture shift

Category conformal policy; normal FPR by family and severity



07 · CONFIRMATORY EVIDENCE

Confirmatory hypothesis tests

Predeclared 95% intervals; positive values support the stated direction



08 · OPERATIONAL BOUNDARY

Local workstation

77.3 ms median · 108.8 ms p95 · 12.55 images/s · 49.97 MB runtime artifacts

Required controls

Known category · verified memory · controlled capture · shift monitoring · visible human review

Decision support only. Heatmaps are feature-space evidence, not physical root-cause diagnosis. No factory trial or quality guarantee is claimed.